American Computer Science League

2020-2021 ● Contest 2: Shorts ● Classroom Division

1. Prefix-Infix-Postfix

Evaluate the following postfix expression if all numbers are single digits:

3 6 * 2 / 3 9 + 4 / − 2 3 ^ +

A. -2
B. 6
C. 12
D. 14
E. 17

2. Prefix-Infix-Postfix

Convert this infix expression to postfix:

$$\frac{A^2 + B}{C} + ABC$$

A. A2^B+/CAB*C*+
B. A2^B+C/AB*C*+
C. A2^B+C/A*BC*+
D. A2B^+C/AB*C*+
E. A2^B+C/ABC**+

3. Prefix-Infix-Postfix

Evaluate the following prefix expression if all numbers are single digits:

− / * 4 + 2 3 * 2 5 / + ^ 6 2 4 8

A. 4
B. 3
C. 0
D. -3
E. -4

4. Prefix-Infix-Postfix

The formula for the surface area of a right circular cylinder is:

$A = 2\pi rh + 2\pi r^2$

Convert the entire formula to prefix using P for π.

A. =+***2Prh**2P^r2A
B. =A+**2Pr*h*2P*^r2
C. =A+***2Prh**2P^r2
D. =A***2Prh+**2P^r2
E. =A+*2P*r*h**2P^r2

5. Bit-String Flicking

Evaluate the following expression:

(01011 OR (NOT 10010) AND 00011)

A. 01011
B. 00101
C. 11011
D. 00011
E. 01010

6. **Bit-String Flicking**

Evaluate the following expression:

(NOT (RSHIFT-2 (LCIRC-2 10011)))

A. 11000
B. 00111
C. 10100
D. 11001
E. 11100

7. Bit-String Flicking

Evaluate the following expression:

((LSHIFT-1 11011) OR (RCIRC-2 01101) AND 01111)

A. 11111
B. 10111
C. 01111
D. 01010
E. 00010

8. Bit-String Flicking

How many bit strings make the following equation TRUE?

((LCIRC-3 X) AND 10110) = 10100

A. 2
B. 3
C. 4
D. 5
E. 8

9. LISP Evaluate the following LISP expression: (ADD (EXP 2 3) (MULT -3 5 (SUB 5 9)) (DIV 18 3))	A. -46 B. 68 C. 74 D. 75 E. 368
10. LISP Evaluate the following LISP expression: (CDR (CAR '((b c d) (e f) g)))	A. (b) B. (d) C. (b c) D. (c d) E. (b d)